Legal Module – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Module:** Legal · **Child Modules:** Legal Documents, Legal Footer and Ownership Notice **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

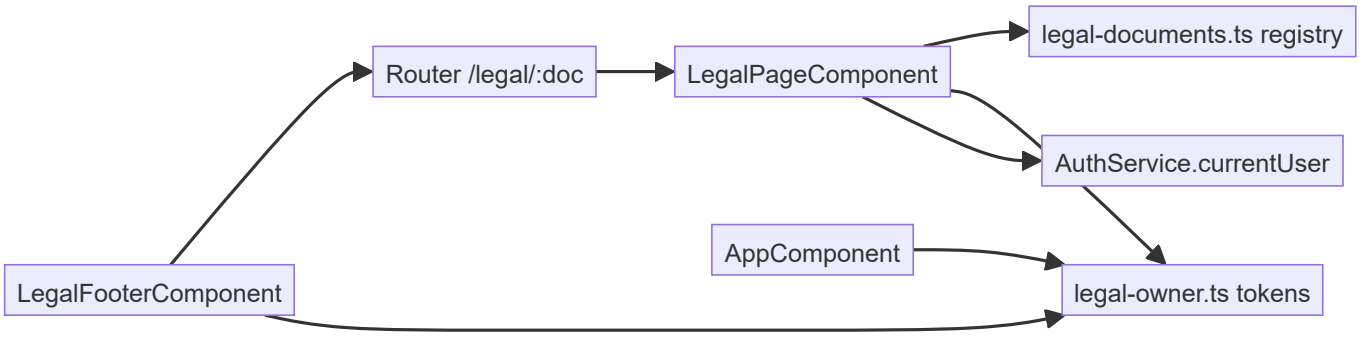
> ✅ Target framework confirmed: `net10.0` (.NET 10), from `backend/src/TaskFlow.Api/TaskFlow.Api.csproj`.

1. Technical Overview

Legal is a **frontend-only** module. It has no controller, service, repository, DTO, entity or table. It contains:

- A **document registry** (`legal-documents.ts`) that holds the five legal documents as typed structured data (`LegalDocument` → `LegalSection` → `LegalBlock`).
- A **single owner record** (`legal-owner.ts`: `LEGAL_OWNER`, `LEGAL_APPLICATION`, `LEGAL_TOKENS`). The registry writes `{{owner}}`, `{{year}}`, `{{email}}`, `{{rights}}`, `{{applicationName}}` and `{{applicationTitle}}` as tokens, and `resolveLegalTokens()` replaces them at render time.
- One standalone renderer, `LegalPageComponent`, used by five lazily loaded public routes (`/legal/copyright|terms|privacy|disclaimer|licenses`). Each route passes `data.documentId`. Detail: `10.1-Legal-Documents.md`.
- A shared `LegalFooterComponent` (`app-legal-footer`, variants `surface` / `dark`). The application shell, the login page and the account recovery page embed it.
- `AppComponent.ngOnInit` sets the document `<title>` and the `author` / `description` / `copyright` meta tags from the same owner record. Detail (footer and metadata): `10.2-Legal-Footer-and-Ownership-Notice.md`.

2. Architecture Flow



No HTTP call is made and no database is involved.

3. Backend File Structure

Layer	Path	Files
Controller	*Not applicable for this module.*	–
Service	*Not applicable for this module.*	–

Layer	Path	Files
Repository	*Not applicable for this module.*	–
DTOs	*Not applicable for this module.*	–
Validators	*Not applicable for this module.*	–
Entities	*Not applicable for this module.*	–
Enums	*Not applicable for this module.*	–
Authorization	*Not applicable for this module.*	–

A search of `backend/src/TaskFlow.Api` for legal / copyright / privacy / terms found no legal endpoint or service. Some backend facts that the Privacy Policy text relies on are listed in section 9 of `10.1-Legal-Documents.md`.

4. Frontend File Structure

Type	Path	Purpose	Child
Module folder	`src/app/modules/legal/` (`legal.routes.ts`, `pages/`, `models/`, `utils/`)	Routes, page component, document model and registry	10.1
Shared owner record	`src/app/modules/legal/utils/legal-owner.ts`	`LEGAL_OWNER`, `LEGAL_APPLICATION`, `LEGAL_TOKENS`, `resolveLegalTokens(text)`, `legalOwnershipLine()`	10.1 and 10.2
Shared component	`src/app/shared/components/legal-footer/`	`LegalFooterComponent` and `LegalFooterVariant`	10.2
Consumers / metadata	`layouts/shell.component.html`, `modules/auth/login/…`, `modules/auth/account-recovery/…`, `app.component.ts`	Embed the footer; set title and meta tags	10.2
Route / Guard	*None*	No guard and no permission check. No entry in `layouts/nav-items.ts`	10.1 and 10.2

The full file lists are in each child document.

5. Child Modules

#	Child Module	Main endpoints (count)	Key permissions	Document
10.1	Legal Documents	0 API endpoints (5 public client routes `/legal/*`)	*(none)*	`10.1-Legal-Documents.md`
10.2	Legal Footer and Ownership Notice	0 API endpoints (no routes; shared component + app metadata)	*(none)*	`10.2-Legal-Footer-and-Ownership-Notice.md`

The child documents are in the `10-Legal/` folder next to this file.

6. Database Design

Not applicable for this module. (No tables, entities or migrations. All content is static frontend data.)

Table	Owned by child	Notes
(none)	–	Both children read only the in-memory `legal-owner.ts` / `legal-documents.ts` constants

Relationships

- *Not applicable for this module.*

7. Security & Authorization

- **Authentication:** Not required. `LEGAL_ROUTES` is registered at root level, outside the shell route that has `authGuard` and `mfaSetupGuard`. `AuthService.currentUser()` is read only to choose the back-link target (10.1).
- **Authorization:** No permission key. `PermissionCatalog.cs` / `PermissionKeys.cs` have no legal key, and the frontend `permissions.ts` and `nav-items.ts` have no legal entry.
- **Data scope:** *Not applicable for this module.* The content is the same for every visitor.
- **Frontend:** No route guard and no permission directive.
- **Output safety:** All document and footer text is rendered with Angular interpolation (`{{ }}`), so it is HTML-escaped. `innerHTML` is not used.

Permission Key	Endpoint(s)	UI Element Controlled
(none)	–	Legal pages and footer are always visible (Always On for every role, and also for visitors who are not signed in)

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
Required field / Duplicate / No permission / Not found (API)	*Not applicable for this module.*	–	–

The global exception middleware does not apply because no API is called. Client-side not-found and unknown-path handling belongs to 10.1 and is described there.

9. Integrations / Side Effects

- **Notifications:** *Not applicable for this module.*
- **Email:** *Not applicable for this module.*
- **Activity Log:** *Not applicable for this module.*
- **AI Assistant:** No tools are exposed. No file in `Services/Assistant/Knowledge/*.md` mentions the legal pages.
- Child-specific side effects (document head metadata in 10.2, Privacy Policy facts checked against the backend in 10.1) are in the child documents.

10. Configuration

Key (`appsettings.json`)	Purpose	Default
None for this module	Owner values are compile-time constants in `legal-owner.ts` (`LEGAL_OWNER.name` "Jojo George", `year` 2026, `email` "jojogeorge3344@gmail.com", `rights` "All Rights Reserved"), used by both children	–

Backend settings that only the Privacy Policy text refers to are listed in `10.1-Legal-Documents.md`.

11. Testing

Test Type	File / Location	Covers
Frontend unit	`src/app/modules/legal/utils/legal-documents.spec.ts`	The registry holds all five documents, each with a title, summary, last-updated date and non-empty sections; required sections per document (Copyright, Terms naming India as governing law, Privacy with no sale/analytics/tracking, Disclaimer, Licences pointing to `THIRD-PARTY-NOTICES.md`); owner, year and email come from `LEGAL_OWNER` as tokens and resolve to the same values as the footer
Frontend unit	`src/app/routes/app.routes.spec.ts`	The five `/legal/*` routes are top-level, unguarded and carry their `documentId`; `/legal/terms` opens `LegalPageComponent` without signing in
Frontend unit	`src/app/modules/legal/pages/legal-page.component.spec.ts`	`LegalPageComponent` renders the document named by the route with one contents entry and one section per document section, replaces every owner token with the `LEGAL_OWNER` values, shows "Legal document not found" for an unknown or missing id, and links back to sign in (signed out) or to the role dashboard (signed in)
Frontend unit	`src/app/shared/components/legal-footer/legal-footer.component.spec.ts`	`LegalFooterComponent` shows the ownership line, links to every document under `/legal/<id>` with its nav label, sets `aria-current="page"` only on the document named by the deepest route, and applies the dark variant on request
Backend	*None*	Legal has no backend code, so there are no backend tests.

12. Known Limitations / Future Enhancements

Module-wide items only. Child-specific gaps are listed in each child document.

Gaps found

- **Public by design:** the legal routes are registered outside the `authGuard` shell, so they need no sign-in and no permission key, and the footer links appear on the login and account recovery screens.
- **Requirements coverage:** Legal is not one of the 20 original screen specs. It is covered by the approved overall BRD (`TF - Documents/TF - BRD/00-Overall/BRD-TaskFlow-Overall.md`, `REQ-LEGAL-01` – `REQ-LEGAL-15`).

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